DGMS 2019 GEOLOGY • QUESTION 39

Phyllosilicates: Muscovite Crystal Chemistry & Layer Stacking

Topic: Mineralogy & Sheet Silicate Crystal Chemistry | Primary Reference: Dexter Perkins (Mineralogy, 3rd ed., Ch. 13 & 14)

Question Statement

Q.39. Muscovite belongs to which one of the following types of phyllosilicate structures? (o = octahedral, t = tetrahedral)

(a) Dioctahedral t-o-t, interlayer cation, dioctahedral t-o-t type
(b) Dioctahedral t-o-t type only
(c) Trioctahedral t-o-t, interlayer cation, trioctahedral t-o-t type
(d) Dioctahedral t-o type only

a **Dioctahedral t-o-t, interlayer cation, dioctahedral t-o-t type (Correct Answer)**

b Dioctahedral t-o-t type only

c Trioctahedral t-o-t, interlayer cation, trioctahedral t-o-t type

d Dioctahedral t-o type only

✓ **Correct Answer: (a) Dioctahedral t-o-t, interlayer cation, dioctahedral t-o-t type [Muscovite possesses 2:1 (t-o-t) sandwiches with 2 out of 3 octahedral sites filled by Al^{3+}, separated by interlayer K^{+} cations].**

Textbook Evidence: Dexter Perkins (Chapter 13 & 14)

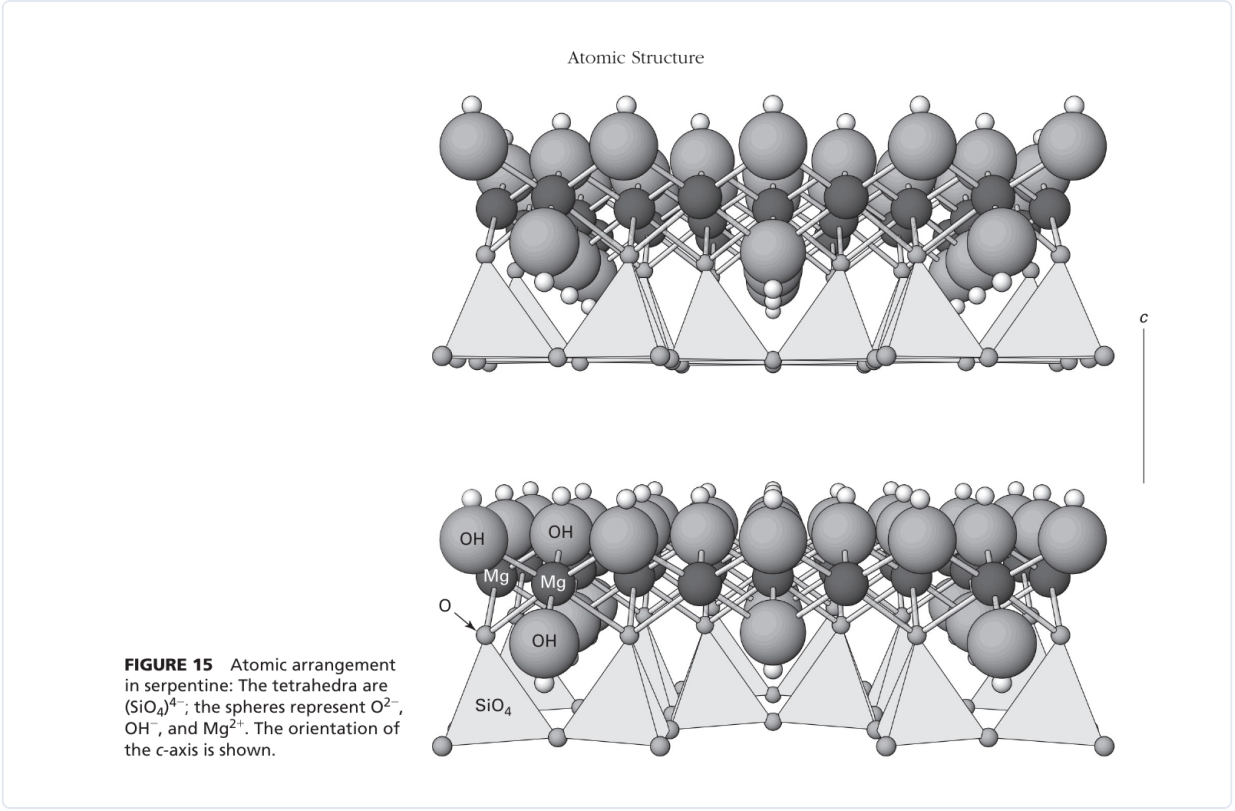


FIGURE 15 Atomic arrangement in serpentine: The tetrahedra are $(SiO_4)^{4-}$; the spheres represent O^{2-}, OH^-, and Mg^{2+}. The orientation of the *c*-axis is shown.

Figure 13.15: Atomic Stacking in Phyllosilicates (Dexter Perkins, *Mineralogy*, 3rd ed., p. 368). In 2:1 sheet silicates (micas, pyrophyllite, talc), an octahedral sheet (o) is sandwiched between two opposing tetrahedral sheets (t), forming a rigid **t-o-t layer**. In *muscovite*, substitution of Al^{3+} for Si^{4+} in tetrahedral sites creates a negative layer charge satisfied by large interlayer K^+ cations situated in 12-fold coordination between successive t-o-t layers.

> *"In pyrophyllite, talc, and micas, tetrahedral layers surround octahedral layers to produce* **T-O-T sandwiches (2:1 layers)***. K^+, and more rarely Na^+ or Ca^{2+}, occupy sites between the sandwiches in micas.*
>
> *When the octahedral sheet contains trivalent aluminum (Al^{3+}), only* **two out of every three octahedral sites** *are occupied to maintain charge balance ($2 \times +3 = +6$); these minerals are* **dioctahedral** *(e.g., muscovite, pyrophyllite, kaolinite). When the octahedral sheet contains divalent cations (Mg^{2+}, Fe^{2+}), all three sites are occupied; these are* **trioctahedral** *(e.g., phlogopite, biotite, talc)."*
>
> **— Dexter Perkins, *Mineralogy*, Chapter 13 (Atomic Structure: Sheet Silicates), pp. 367–368**

Comprehensive Option Analysis

(a) Dioctahedral t-o-t, interlayer cation, dioctahedral t-o-t type **[CORRECT]**

Crystal Chemistry of Muscovite: The structural formula of muscovite is $\text{K}\text{Al}_2[\text{AlSi}_3\text{O}_{10}](\text{OH})_2$:

- **Layer Architecture (2:1 or t-o-t):** One octahedral sheet (o) is bonded between two inverted tetrahedral sheets (t).
- **Dioctahedral Nature:** Aluminum (Al^{3+}) resides in the octahedral sheet. To balance charge, only 2 out of 3 octahedral positions per formula unit are filled by Al^{3+}, leaving 1 site vacant ($2/3$ occupancy $\equiv$ **dioctahedral**).
- **Interlayer Cation:** In the tetrahedral sheets, one out of four Si^{4+} cations is replaced by Al^{3+} (AlSi_3 instead of Si_4). This generates a net -1 electrostatic deficit per formula unit, which is neutralized by **interlayer K^+ cations** located between adjacent t-o-t layers.
- Therefore, the full stacking architecture is: **dioctahedral t-o-t → interlayer cation (K^+) → dioctahedral t-o-t**.

(b) Dioctahedral t-o-t type only **[INCORRECT]**

This describes **pyrophyllite** ($\text{Al}_2\text{Si}_4\text{O}_{10}(\text{OH})_2$). In pyrophyllite, all tetrahedral sites are occupied by Si^{4+} without substitution, leaving the t-o-t layers electrically neutral. As a result, *no interlayer cations exist*; the t-o-t sheets are held together strictly by weak van der Waals forces, making pyrophyllite very soft (hardness 1–2).

(c) Trioctahedral t-o-t, interlayer cation, trioctahedral t-o-t type **[INCORRECT]**

This describes the **trioctahedral micas**—namely **biotite** ($\text{K}(\text{Mg}, \text{Fe})_3[\text{AlSi}_3\text{O}_{10}](\text{OH})_2$) and **phlogopite** ($\text{K}\text{Mg}_3[\text{AlSi}_3\text{O}_{10}](\text{OH})_2$). Because they contain divalent cations ($\text{Mg}^{2+}, \text{Fe}^{2+}$), all 3 out of 3 octahedral sites are occupied ($3/3 \equiv$ trioctahedral).

(d) Dioctahedral t-o type only **[INCORRECT]**

This describes **kaolinite** ($\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4$), a 1:1 clay mineral consisting of alternating single tetrahedral (t) and single octahedral (o) sheets without interlayer cations.

Master Structural Taxonomy of Phyllosilicates

Layer Type	Octahedral Character	Interlayer Contents	Diagnostic Mineral	Chemical Formula
1:1 (t-o)	Dioctahedral (Al^{3+} in 2/3 sites)	None (Hydrogen bonding)	**Kaolinite**	$\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4$
	Trioctahedral (Mg^{2+} in 3/3 sites)	None (Hydrogen bonding)	**Serpentine (Antigorite/ Chrysotile)**	$\text{Mg}_3\text{Si}_2\text{O}_5(\text{OH})_4$
2:1 (t-o-t)	Dioctahedral (Al^{3+} in 2/3 sites)	None (Van der Waals bonds)	**Pyrophyllite**	$\text{Al}_2\text{Si}_4\text{O}_{10}(\text{OH})_2$
	Trioctahedral (Mg^{2+} in 3/3 sites)	None (Van der Waals bonds)	**Talc**	$\text{Mg}_3\text{Si}_4\text{O}_{10}(\text{OH})_2$
	Dioctahedral (Al^{3+} in 2/3 sites)	**Interlayer Cation (K^+)**	**Muscovite (True White Mica)**	**$\text{K}\text{Al}_2[\text{AlSi}_3\text{O}_{10}](\text{OH})_2$**
	Trioctahedral ($\text{Mg}^{2+}/\text{Fe}^{2+}$ in 3/3 sites)	Interlayer Cation (K^+)	**Phlogopite / Biotite**	$\text{K}(\text{Mg},\text{Fe})_3[\text{AlSi}_3\text{O}_{10}](\text{OH})_2$
2:1:1 (t-o-t + o)	Trioctahedral / Dioctahedral	Interlayer brucite-like octahedral sheet	**Chlorite group**	$(\text{Mg},\text{Fe})_5\text{Al}(\text{AlSi}_3\text{O}_{10})(\text{OH})_8$

Key Takeaway & Exam Strategy

Mnemonic & Structural Classification Rule:

- **Muscovite = Dioctahedral** (Al^{3+}, 2 out of 3 sites filled) + **2:1 (t-o-t)** + **Interlayer K^+**.

- **Biotite** / **Phlogopite = Trioctahedral** (Mg^{2+}/Fe^{2+}, 3 out of 3 sites filled) + **2:1 (t-o-t)** + **Interlayer K^{+}**.
- **Pyrophyllite** / **Talc = Neutral t-o-t** (no interlayer cation, soft).
- **Kaolinite = 1:1 (t-o) dioctahedral**.